

Long-Term Option Strategies

Long-term option strategies are very similar to those involving short-term options, but there *are* some nuances. Listed long-term options are actually a slightly different class of options called LEAPS. While that term still exists, it is somewhat outmoded. Rather, in the modern vernacular, long-term options are just referred to by their month and year. For example, in the year 2012, one might refer to an “IBM Jan ‘14 call”—one expiring in January 2014.

LEAPS options were first introduced by the CBOE in October 1990, and were offered on a handful of blue-chip stocks. They proved quite popular, and eventually long-term options were listed on more stocks. Today, they exist on many stocks and indices with listed options. Prior to the Options Symbol Initiative (OSI) (see Chapter 1), an entirely different set of symbols was required to describe them. But once the OSI came into effect in 2010, the term LEAPS was no longer mandatory.

Strategies involving long-term options are not substantially different from those involving shorter-term options. However, the fact that the option has so much time remaining seems to favor the buyer and be a detriment to the seller. This is one reason why LEAPS have been popular. As a strategist, one knows that the length of time remaining has little to do with whether a certain strategy makes sense or not. Rather, it is the relative value of the option that dictates strategy. If an option is overpriced, it is a viable candidate for selling, whether it has two years of life remaining or two months. Obviously, follow-up action may become much more of a necessity during the life of a two-year option; that matter is discussed later in this chapter.

Long-term options are generally listed about 2.5 years before they expire. For example, in the fall of 2011, long-term options expiring in January of 2014 were listed on most stocks. These listings used to occur in the spring, meaning that the longest-term options had about 2.75 years until expiration, but in 2008, the listed period was shortened. For long-term *stock* options, the only expiration month is January. However, for some index options, long-term options expire in December instead. The details stated above in this paragraph could easily change in the future.

PRICING LEAPS

The factors influencing the prices of LEAPS are the same as those for any other option:

1. underlying stock price,
2. striking price,
3. time remaining,
4. volatility,
5. risk-free interest rate, and
6. dividend rate.

The relative influence of these factors may be a little more pronounced for LEAPS than it is for shorter-term equity options. Consequently, the trader may think that a LEAPS is overly expensive or cheap by inspection, when in reality it is not. *One should be careful in his evaluation of LEAPS until he has acquired experience in observing how their prices relate to the shorter-term equity options with which he is experienced.*

It might prove useful to reexamine the option pricing curve with some LEAPS included. Please refer to Figure 25-1 for the pricing curves of several options. As always, the solid intrinsic value line is the bottom line; it is the same for any call option. The curves are all drawn with the same values for the pertinent variables: stock price, striking price, volatility, short-term interest rate, and dividends. Thus, they can be compared directly.

The most obvious thing to notice about the curves in Figure 25-1 is that the curve depicting the 2-year LEAPS is quite flat. It has the *general* shape of the shorter-term curves, but there is so much time value at stock prices even 25% in- or out-of-the-money, that the 2-year curve is much flatter than the others.

Other observations can be made as well. Notice the at-the-money options: The 2-year LEAPS sells for a little more than four times the 3-month option. As we shall see,

FIGURE 25-1.
Call option pricing curve.



this can change with the effects of interest rates and dividends, but it confirms something that was demonstrated earlier: Time decay is not linear. Thus, the 2-year LEAPS, which has eight times the amount of time remaining as compared to the 3-month call, only sells for about four times as much. This LEAPS might appear cheap to the casual observer, but remember that these graphs depict the fair values for this set of input parameters. *Do not be deluded into thinking that a LEAPS looks cheap merely by comparing its price to a nearer-term option; use a model to evaluate it, or at least use the output of someone else's model.*

The curves in Figure 25-1 depict the relationships between stock price, striking price, and time remaining. The most important remaining determinant of an option's price is the volatility of the underlying stock. Changes in volatility can greatly change the price of any option. This is especially true for LEAPS, since a long-term option's price will fluctuate greatly when volatility changes only a little. Some observations on the differing effects that volatility changes have on short- and long-term options are presented later.

Before that discussion, however, it may be beneficial to examine the effects that interest rates and dividends can have on LEAPS. These effects are much, much greater than those on conventional equity options. Recall that it was stated that interest rates and dividends are minor determinants in the price of an option, unless the dividends were large. That statement pertains mostly to short-term options. For longer-term options, the cumulative effect of an interest rate or dividend over such a long period of time can have a magnified effect in terms of the absolute price of the option.

FIGURE 25-2.
2-year call pricing curve, interest rate comparison.

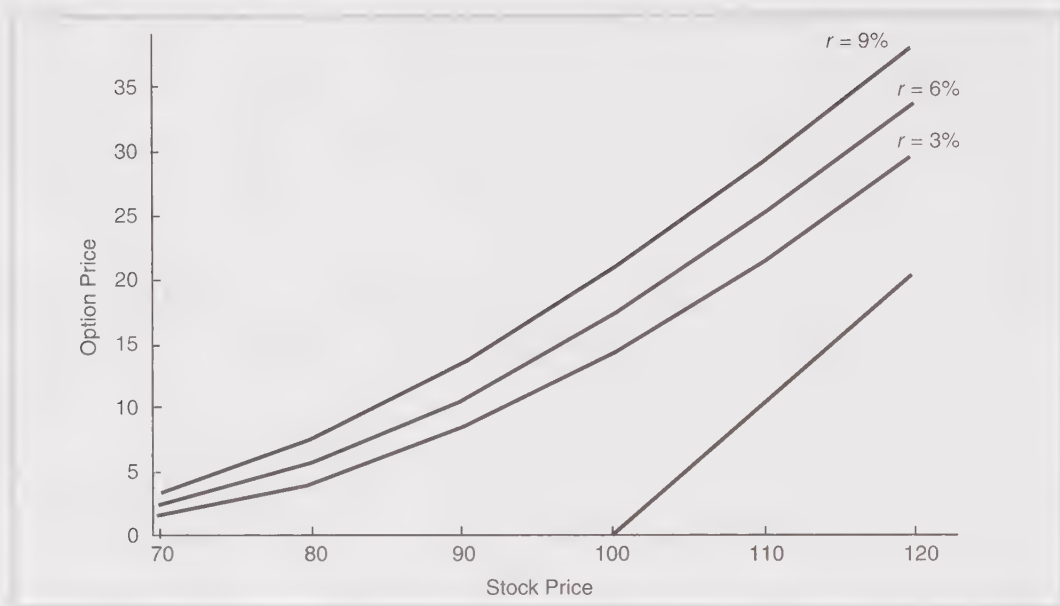
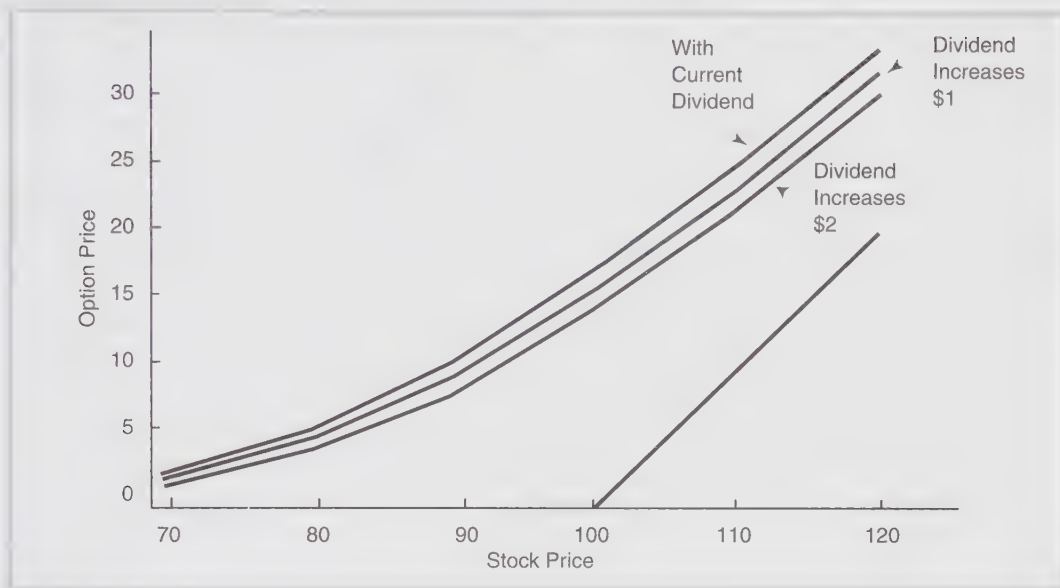


Figure 25-2 presents the option pricing curve again, but the only option depicted is a 2-year call option. The striking price is 100, and the straight line at the right depicts the intrinsic value. The three curves represent option prices for risk-free interest rate of 3%, 6%, and 9%. All other factors (time to expiration, volatility, and dividends) are fixed. The difference between option prices caused merely by a shift in rates of 3% is very large.

The difference in LEAPS prices increases as the LEAPS becomes in-the-money. Note that in this figure, the distance between the curves gets wider as one scans them from left to right. The price difference for *out-of-the-money* LEAPS is large enough—nearly a point even for options fairly far out-of-the-money (that is, the points on the left-hand side of the graph). A shift of 3% in rates causes a larger price difference of over 2 points in the *at-the-money*, 2-year LEAPS. The largest differential in option prices occurs *in-the-money*! This may seem somewhat illogical, but when LEAPS strategies are examined later, the reasons for this will become clear. Suffice it to say that the in-the-money LEAPS are changed in price by over 4 points when rates change by 3%. That is a monstrous differential and should cause any trader who is considering trading in-the-money LEAPS to consider what his outlook is for short-term interest rates.

There is always a substantial probability that rates can change by 3% in two years. Thus, it is difficult to predict with any certainty what risk-free rate to use in the pricing of

FIGURE 25-3.
Long-term call pricing curve as dividends increase.



two-year LEAPS. Moreover, one should be very careful when deciding LEAPS are “cheap” or “expensive” because, conventionally, the short-term interest rate is not usually considered as a significant factor in making such an analysis. For LEAPS, however, Figure 25-2 is obvious proof that interest rate considerations are important.

Now consider dividends. Figure 25-3 depicts the prices of two-year calls. The three curves on the graph are for different dividend rates—the top line representing the current rate, the middle line representing prices if the dividend were raised by \$1 annually, and the bottom line showing what prices would be if the dividend were raised by \$2 annually. All other factors (volatility, time remaining, and risk-free interest rates) are the same for each curve in this graph. The increase in dividends manifests itself by decreasing the LEAPS call price. The reason that this is true, of course, is that the stock will be reduced in price more when it goes ex-dividend by the larger amounts of the increased dividends.

The actual amount that the calls lose in price increases slightly as the call is more in-the-money. That is, the curves are closer together on the left-hand (out-of-the-money) side than they are on the right-hand (in-the-money) side. For the in-the-money call, a \$1 increase in dividends over two years can cause the LEAPS to be worth about $1\frac{1}{2}$ points less in value.

Figure 25-3 is to the same scale as Figure 25-2, so they can be compared directly in terms of magnitude. Notice that the effect of a \$1 increase in dividends on the call

prices is much smaller than that of an increase in interest rates by 3%. Graphically speaking, one can observe this by noting that the spaces between the three curves in the previous figure are much wider than the spaces between the three curves in this figure.

Finally, note that dividend increases have the opposite effect on puts. That is, an increase in the dividend payout of the underlying common will cause a put to *increase* in price. If the put is a long-term put, then the effect of the increase will be even larger.

Lest one think that LEAPS are too difficult to price objectively, note the following. The prior figures of interest rate and dividend effects tend to magnify the effects on prices for two reasons. First, they depict the effects on 2-year LEAPS. That is a large amount of life. The effects would be diminished somewhat for options with 10 to 23 months of life left. Second, the figures depict the change in rates or dividends as being instantaneous. This is not completely realistic. If rates change, they will change by a little bit at a time, usually $\frac{1}{4}\%$ or $\frac{1}{2}\%$ at a time, perhaps as much as 1%. If dividends are increased, that increase may be instantaneous, but it will not likely occur immediately after the LEAPS are purchased or sold. However, the point that these figures are meant to convey is that interest rates and dividends have a much greater effect on LEAPS than on ordinary shorter-term equity options, and that is certainly a true statement.

COMPARING LEAPS AND SHORT-TERM OPTIONS

Table 25-1 will help to illustrate the problem in valuing LEAPS, either mentally or with a model. All of the variables—stock price, volatility, interest rates, and dividends—are given in increments and the comparison is shown between 3-month equity options and 2-year LEAPS. There are three sets of comparisons: for options 20% out-of-the-money, options at-the-money, and options 20% in-the-money.

TABLE 25-1.
Comparing LEAPS and Short-Term Calls.

Variable	Increment	Change in Price of the Options					
		20% out		at		20% in	
		3-mo.	2-yr.	3-mo.	2-yr.	3-mo.	2-yr.
Stock Prc.	+ 1pt	.03	.41	.54	.70	.97	.89
Volatility	+ 1%	.03	.43	.21	.48	.04	.33
Int. Rate	+ $\frac{1}{2}\%$	.01	.27	.08	.55	.14	.72
Dividend	+ \$.25/qtr	0	-.62	-.08	-1.18	-.14	-1.50

A few words are needed here to explain how volatility is shown in this table. Volatility is normally expressed as a percent. The volatility of the stock market is about 15%. The table shows what would happen if volatility changed by one percentage point, to 16%, for example. Of course, the table also shows what would happen if the other factors changed by a small amount.

Most of the discrepancies between the 3-month and the 2-year options are large. For example, if volatility increases by one percentage point, the 3-month out-of-the-money call will increase in price by only 3 cents (0.03 in the left-hand column) while the 2-year LEAPS call will increase by 43 cents. As another example, look at the bottom right-hand pair of numbers, which show the effect of a dividend increase on the options that are 20% in-the-money. The assumption is that the dividend will increase 25 cents this quarter (and will be 25 cents higher every quarter thereafter). This translates into a loss of 14 cents for the 3-month call, since there is only one ex-dividend period that affects this call; but it translates into a loss of 1½ for the 2-year LEAPS, since the stock will go ex-dividend by an extra \$2 over the life of that call.

The table also shows that only three of the discrepancies are not large. Two involve the stock price change. If the stock changes in price by 1 point, neither the at-the-money nor the in-the-money options behave very differently, although the at-the-money LEAPS do jump by 70 cents. The observant reader will notice that the top line of the table depicts the *delta* of the options in question; it shows the change in option price for a one-point change in stock price. The only other comparison that is not extremely divergent is that of volatility change for the at-the-money option. The 3-month call changes by 21 cents while the LEAPS changes by nearly ½ point. This is still a factor of two-to-one, but is much less than the other comparisons in the table.

Study the other comparisons in the table. The trader who is used to dealing with short-term options might ordinarily ignore the effect of a rise in interest rates of ½ of 1%, of a 25-cent increase in the quarterly dividend, of the volatility increasing by a mere 1%, or maybe even of the stock moving by one point (only if his option is out-of-the-money). The LEAPS option trader will gain or suffer substantially and immediately if any of these occur. In almost every case, his LEAPS call will gain or lose ½ point of value—a significant amount, to be sure.

LEAPS STRATEGIES

Many of the strategies involving LEAPS are not significantly different from their counterparts that involve short-term options. However, as shown earlier, the long-term nature of the LEAPS can sometimes cause the strategist to experience a result different from that to which he has become accustomed.

As a general rule, one would want to be a buyer of LEAPS when interest rates were low and when the volatilities being implied in the marketplace are low. If the opposite were true (high rates and high volatilities), he would lean toward strategies in which the sale of LEAPS is used. Of course, there are many other specific considerations when it comes to operating a strategy, but since the long-term nature of LEAPS exposes one to interest rate and volatility movements for such a long time, one may as well attempt to position himself favorably with respect to those two elements when he enters a position.

LEAPS AS STOCK SUBSTITUTE

Any in-the-money option can be used as a substitute for the underlying stock. Stock owners may be able to substitute a long in-the-money call for their long stock. Short sellers of stock may be able to substitute a long put for their short stock. This is not a new idea; it was discussed briefly in Chapter 3 under reasons why people buy calls. It has been available as a strategy for some time with short-term options. Its attractiveness seems to have increased somewhat with the introduction of LEAPS, however. More and more people are examining the potential of selling the stock they own and buying long-term calls (LEAPS) as a substitute, or buying LEAPS instead of making an initial purchase in a particular common stock.

Substitution for Stock Currently Held Long. Simplistically, this strategy involves this line of thinking: If one owns stock and sells it, an investor could reinvest a small portion of the proceeds in a call option, thereby providing continued upside profit potential if the stock rises in price, and invest the rest in a bank to earn interest. The interest earned would act as a substitute for the dividend, if any, to which the investor is no longer entitled. Moreover, he has less downside risk: If the stock should fall dramatically, his loss is limited to the initial cost of the call.

In actual practice, one should carefully calculate what he is getting and what he is giving up. For example, is the loss of the dividend too great to be compensated for by the investment of the excess proceeds? How much of the potential gain will be wasted in the form of time value premium paid for the call? The costs to the stock owner who decides to switch into call options as a substitute are commissions, the time value premium of the call, and the loss of dividends. The benefits are the interest that can be earned from freeing up a substantial portion of his funds, plus the fact that there is less downside risk in owning the call than in owning the stock.

Example: XYZ is selling at 50. There are one-year LEAPS with a striking price of 40 that sell for \$12. XYZ pays an annual dividend of \$0.50 and short-term interest rates are 5%. What are the economics that an owner of 100 XYZ common stock must calculate in

order to determine whether it is viable to sell his stock and buy the one-year LEAPS as a substitute?

The call has time value premium of 2 points ($40 + 12 - 50$). Moreover, if the stock is sold and the LEAPS purchased, a credit of \$3,800 less commissions would be generated. First, calculate the net credit generated:

Credit balance generated:

Sale of 100 XYZ stock	\$5,000
Less stock commission	<u>25</u>
Net sale proceeds:	\$4,975 credit

Cost of one LEAPS call	\$1,200
Plus option commission	<u>15</u>
Net cost of call:	\$1,215 debit
Total credit balance:	\$3,760 credit

Now the costs and benefits of making the switch can be computed:

Costs of switching:

Time value premium	– \$200
Loss of dividend	– \$ 50
Stock commissions	– \$ 25
Option commissions	<u>– \$ 15</u>
Total cost:	– \$290

Fixed benefit from switching:

Interest earned on	
credit balance of \$3,760	
at 5% interest for one year = $0.05 \times \$3,760$;	+ \$188

Net cost of switching:	– \$102
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The stock owner must now decide if it is worth just over \$1 per share in order to have his downside risk limited to a price of $39\frac{1}{2}$ over the next year. The price of $39\frac{1}{2}$ as his downside risk is merely the amount of the net credit he received from doing the switch (\$3,760) plus the interest earned (\$188), expressed in per-share terms. That is, if XYZ falls dramatically over the next year and the LEAPS expires worthless, this investor will still

have \$3,948 in a bank account. That is equivalent to limiting his risk to about 39½ on the original 100 shares.

If the investor decides to make the substitution, he should invest the proceeds from the sale in a 1-year CD or Treasury bill, for two reasons. First, he locks in the current rate—the one used in his calculations—for the year. Second, he is not tempted to use the money for something else, an action that might negate the potential benefits of the substitution.

The above calculations all assume that the LEAPS call or the stock would have been held for the full year. If that is known not to be the case, the appropriate costs or benefits must be recalculated.

Caveats. This (\$102) seems like a reasonably small price to pay to make the switch from common stock to call ownership. However, if the investor were planning to sell the stock before it fell to 39½ in any case, he might not feel the need to pay for this protection. (Be aware, however, that he could accomplish essentially the same thing, since he can sell his LEAPS call whenever he wants to.) Moreover, when the year is up, he will have to pay another stock commission to repurchase his XYZ common if he still wants to own it (or he will have to pay two option commission to roll his long call out to a later expiration date). One other detriment that might exist, although a relatively unlikely one, is that the underlying common might declare an increased dividend or, even worse, a special cash dividend. The LEAPS call owner would not be entitled to that dividend increase in whatever form, while, obviously, the common stock owner would have been. If the company declared a stock dividend, it would have no effect on this strategy since the call owner is entitled to those. A change in interest rates is not a factor either, since the owner of the LEAPS should invest in a 1-year Treasury bill or a 1-year CD and therefore would not be subject to interim changes in short-term interest rates.

There may be other mitigating circumstances. Mostly these would involve tax considerations. If the stock is currently a profitable investment, the sale would generate a capital gain, and taxes might be owed. If the stock is currently being held at a loss, the purchase of the call would constitute a wash sale and the loss could not be taken at this time. (See Chapter 41 on taxes for a broader discussion of the wash sale rule and option trading.)

In theory, the calculations above could produce an overall credit, in which case the stockholder would normally want to substitute with the call, unless he has overriding tax considerations or suspects that a cash dividend increase is going to be announced. *Be very careful about switching if this situation should arise.* Normally, arbitrageurs—persons trading for exchange members and paying no commission—would take advantage of such a situation before the general public could. If they are letting the opportunity pass by, there must be a reason (probably the cash dividend), so be extremely certain of your economics and research before venturing into such a situation.

In summary, holders of common stock on which there exist in-the-money LEAPS should evaluate the economics of substituting the LEAPS call for the common stock. Even if arithmetic calculations call for the substitution, the stockholder should consider his tax situation as well as his outlook for the cash dividends to be paid by the common before making the switch.

BUYING LEAPS AS THE INITIAL PURCHASE INSTEAD OF BUYING A COMMON STOCK

Logic similar to that used earlier to determine whether a stockholder might want to substitute a LEAPS call for his stock can be used by a prospective purchaser of common stock. In other words, this investor does not already own the common. He is going to buy it. This prospective purchaser might want to buy a LEAPS call and put the rest of the money he had planned to use in the bank, instead of actually buying the stock itself.

His costs—real and opportunity—are calculated in a similar manner to those expressed earlier. The only real difference is that he has to spend the stock commission in this case, whereas he did not in the previous example (since he already owned the stock).

Example: As before, XYZ is selling at 50; there are 1-year LEAPS with a striking price of 40 that sell for \$12; XYZ pays an annual dividend of \$0.50, and short-term interest rates are 5%.

The *initial* purchaser of common stock would have certain “opportunity” costs and savings if he decided instead to buy the LEAPS calls. First, calculate the difference in investment required for the stock versus the LEAPS:

Prospective initial investment:			
Stock:	$\$5,000 + \25 commission	=	\$5,025
LEAPS:	$\$1,200 + \15 commission	=	\$1,215
Net difference:			\$3,810

Now calculate the costs versus the savings:

Costs:	
Time value premium	–\$200
Loss of dividend	–\$ 50
Savings:	
Interest on \$3,810 for one year at 5%:	+\$190
Net opportunity cost:	–\$ 60

In this case, it seems even more likely that the prospective stock purchaser would instead buy the LEAPS call. His net “cost” of doing so, provided he puts the difference in initial investment in a 1-year CD or Treasury bill, is only \$60. For this small amount, he has all the upside appreciation (except \$60 worth), but has risk only down to 40 (he will have \$4,000 in his bank account at the end of one year even if the LEAPS expire worthless).

This strategy of buying in-the-money LEAPS and putting the difference between the LEAPS cost and the stock cost in an interest-bearing instrument is an attractive one. It might seem it would be especially attractive if interest rates for the differential were high. Unfortunately, those high rates would present something of a catch-22 because, as was shown earlier, higher rates will cause the LEAPS to be more expensive.

In this margin strategy, one has the risk of not participating in cash dividend increases or specials as the stockholder who substitutes does. But the other concerns of the stockholder, such as taxes, are not pertinent here. Again, these specific calculations only apply if the stock were to be held for the entire year. Adjustments would have to be made if the holding period envisioned is shorter.

Using Margin. The same prospective initial purchaser of common stock might have been contemplating the purchase of the stock on margin. If he used the LEAPS instead, he could save the margin interest. Of course, he wouldn’t have as much money to put in the bank, but he should also compare his costs against those of buying the LEAPS call instead.

Example: As before, XYZ is selling at 50; there are 1-year LEAPS with a striking price of 40 that sell for \$12; XYZ pays an annual dividend of \$0.50; and short-term interest rates are 5%. Furthermore, assume the margin rate is 8% on borrowed debit balances.

First, calculate the difference in prospective investments:

Cost of buying the stock:	
\$5,000 + \$25 commission:	\$5,025
Amount borrowed (50%)	-2,512
Equity required	\$2,513
Cost of buying LEAPS:	
\$1,200 + \$15 commission:	\$1,215
Difference (available to be placed in bank account)	\$1,298

Now the costs and opportunities can be compared, if it is assumed that he buys the LEAPS:

Costs:

Time value premium	-\$200
Dividend loss	- 50

Savings:

Interest on \$1,298 at 5%	+\$ 65
Margin interest on \$2,512 debit balance at 8% for one year	+ 201

Net Savings:	+\$ 16
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For the prospective margin buyer, there is a real savings in this example. The fact that he does not have to pay the margin interest on his debit balance makes the purchase of the LEAPS call a cost-saving alternative. Finally, it should be noted that current margin rules allow one to purchase a LEAPS option on margin. That can be accounted for in the above calculations as well; merely reduce the investment required and increase the margin charges on the debit balance.

In summary, a prospective purchaser of common stock may often find that if there is an in-the-money option available, the purchase of that option is more attractive than buying the common stock itself. If he were planning to buy on margin, it is even more likely that the LEAPS purchase will be attractive. The main drawback is that he will not participate if cash dividends are increased or a special dividend is declared. Read on, however, because the next strategy may be better than the one above.

PROTECTING EXISTING STOCK HOLDINGS WITH LEAPS PUTS

What was accomplished in the substitution strategy previously discussed? The stock owner paid some cost (\$102 in the actual example) in order to limit the risk of his stock ownership to a price of 39½. What if he had bought a LEAPS put instead? Forgetting the price of the put for a moment, concentrate on what the strategy would accomplish. He would be protected from a large loss on the downside since he owns the put, and he could participate in upside appreciation since he still owns the stock. Isn't this what the substitution strategy was trying to accomplish? *Yes, it is.* In this strategy, only one commission is paid—that being on a fairly cheap out-of-the-money LEAPS put—and there is no risk of losing out on dividend increases or special dividends.

The comparison between substituting a call or buying a put is a relatively simple one. First, do the calculations as they were performed in the initial example above. That

example showed that the stockholder's cost would be \$102 to substitute the LEAPS call for the stock, and such a substitution would protect him at a price of $39\frac{1}{2}$. In effect, he is paying \$152 for a LEAPS put with a strike of 40—the \$102 cost plus the difference between 40 and the $39\frac{1}{2}$ protection price. Now, if an XYZ 1-year LEAPS put with strike 40 were available at $1\frac{1}{2}$, he could accomplish everything he had initially wanted merely by buying the put.

Moreover, he would save commissions and still be in a position to participate in increased cash dividends. These additional benefits should make the put worth even more to the stockholder, so that he might pay even slightly more than $1\frac{1}{2}$ for the put. If the LEAPS put were available at this price, it would clearly be the better choice and should be bought instead of substituting the LEAPS call for the common stock.

Thus, any stockholder who is thinking of protecting his position can do it in one of two ways: Sell the stock and substitute a call, or continue to hold his stock and buy a put to protect it. LEAPS calls and puts are amenable to this strategy. Because of the LEAPS' long-term nature, one does not have to keep reestablishing his position and pay numerous commissions, as he would with short-term options. The stockholder should perform the simple calculations as shown above in order to decide whether the move is feasible at all, and if it is, whether to use the call substitution strategy or the put protection strategy.

LEAPS INSTEAD OF SHORT STOCK

Just as in-the-money LEAPS calls may sometimes be a smarter purchase than the stock itself, in-the-money puts may sometimes be a better purchase than *shorting* the common stock. Recall that either the put purchase or the short sale of stock is a bearish strategy, generally implemented by someone who expects the stock to decline in price. The strategist knows, however, that short stock is a component of many strategies and might reflect other opinions than pure bearishness on the common. In any case, an in-the-money put may prove to be a viable substitute for shorting the stock itself. The two main advantages that the put owner has are that he has limited risk (whereas the short seller of stock has theoretically unlimited risk); and he does not have to pay out any dividends on the underlying stock as the short seller would. Also, the commissions for buying the put would normally be smaller than those required to sell the stock short.

There is not much in the way of calculating that needs to be done in order to make the comparison between buying the in-the-money put and shorting the stock. If the time value premium spent is small in comparison with the dividend payout that is saved, then the put is probably the better choice.

Professional arbitrageurs and other exchange members, as well as some large customers, receive interest on their short sales. For these traders, the put would have to be trading with virtually no time premium at all in order for the comparison to favor the put

purchase over the stock short sale. However, the public customer who is going to be shorting stock should be aware of the potential for buying an in-the-money put instead.

SPECULATIVE OPTION BUYING WITH LEAPS

Strategists know that buying calls and puts can have various applications; witness the stock substitution strategies above. However, the most popular reason for buying options is for speculative gain. The leverage inherent in owning options and their limited risk feature make them attractive for this purpose as well. The risk, of course, can be 100% of the investment, and time decay works against the option owner as well. LEAPS calls and puts fit all of these descriptions; they simply have longer maturities.

Time decay is the major enemy of the speculative option holder. Purchasing LEAPS options instead of the shorter-term equity options generally exposes the buyer to less risk of time decay on a daily basis. This is true because the extreme negative effects of time decay magnify as the option approaches its expiration. Recall that it was shown in Chapter 3 that time decay is not linear: An option decays more rapidly at the end of its life than at the beginning. Eventually, a LEAPS put or call will become a normal short-term equity option and time will begin to take a more rapid toll. But in the beginning of the life of LEAPS, there is so much time remaining that the short-term decay is not large in terms of price.

Table 25-2 and Figure 25-4 depict the rate of decay of two options: one is at-the-money (the lower curve) and the other is 20% out-of-the-money (the upper curve). The horizontal axis is months of life remaining until expiration. The vertical axis is the percent of the option price that is lost *daily* due to time decay. The options that qualify as LEAPS are ones with more than 9 months of life remaining, and would thus be the ones on the lower right-hand part of the graph.

The upward-sloping nature of both curves as time to expiration wanes shows that time decay increases more rapidly as expiration approaches. Notice how much more rapidly the out-of-the-money option decays, percentagewise, than the at-the-money. LEAPS, however, do not decay much at all compared to normal equity options. Most LEAPS, even the out-of-the-money ones, lose less than $\frac{1}{4}$ of one percent of their value daily. This is a pittance when compared with a 6-month equity option that is 20% out-of-the-money—that option loses well over 1% of its value daily and it still has 6 months of life remaining.

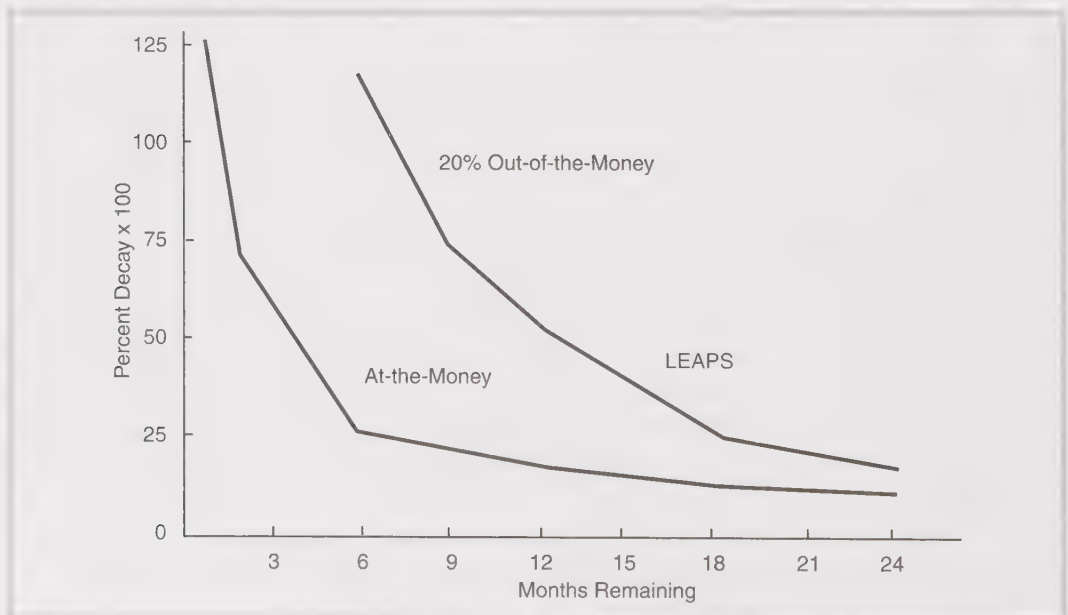
From the accompanying table, observe that the out-of-the-money 2-month option loses over 4% of its value daily!

Thus, LEAPS do not decay at a rapid rate. This gives the LEAPS holder a chance to have his opinion about the stock price work for him without having to worry as much

TABLE 25-2.
Daily percent time value decay.

Months remaining	Percent Decay	
	At-the-money	20% Out-of-the-money
24	.12	.18
18	.14	.27
12	.19	.55
9	.22	.76
6	.27	1.18
3	.60	3.57
2	.73	4.43
1	1.27	
2 wks	3.33	

FIGURE 25-4.
Daily percent time value decay.



about the passage of time as the average equity option holder would. An advantage of owning LEAPS, therefore, is that one's timing of the option purchase does not have to be as exact as that for shorter-term option buying. This can be a great psychological advantage as well as a strategic advantage. The LEAPS option buyer who feels strongly that the stock will move in the desired direction has the luxury of being able to wait calmly for the anticipated move to take place. If it does not, even in perhaps as long as 6 months' time, he may still be able to recoup a reasonable portion of his initial purchase price because of the slow percentage rate of decay.

Do not be deluded into believing that LEAPS don't decay at all. Although the *rate* of decay is slow (as shown previously), an option that is losing 0.15% of its value daily will still lose about 25% of its value in six months.

Example: XYZ is at 60 and there are 18-month LEAPS calls selling for \$8, with a striking price of 60. The daily decay of this call with respect to time will be minuscule; it will take about a week for even an eighth of a point to be lost due to time. However, if the option is held for six months and nothing else happens, the LEAPS call will be selling for about 6. Thus, it will have lost 25% of its value if the stock remains around 60 at the end of six months.

Those familiar with holding equity calls and puts are more accustomed to seeing an option lose 25% of its value in possibly as little as four or five weeks' time. Thus, the advantage of holding the LEAPS is obvious from the viewpoint of slower time decay.

This observation leads to the obvious question: "When is the best time to sell my call and repurchase a longer-term one?" Referring again to Figure 25-4 may help answer the question. Note that for the at-the-money option, the curve begins to bend dramatically upward soon after the 6-month time barrier is passed. Thus, it seems logical that to minimize the effects of time decay, all other things being equal, one would sell his long at-the-money call when it has about 6 months of life left and simultaneously buy a 2-year LEAPS call. This keeps his time decay exposure to a minimum.

The out-of-the-money call is more radical. Figure 25-4 shows that the call that is 20% out-of-the-money begins to decay much more rapidly (percentagewise) at sometime just before it reaches one year until expiration. The same logic would dictate, then, that if one is trading out-of-the-money options, he would sell his option held long when it has about one year to go and reestablish his position by buying a 2-year LEAPS option at the same time.

ADVANTAGES OF BUYING "CHEAP"

It has been demonstrated that rising interest rates or rising volatility would make the price of a LEAPS call increase. Therefore, if one is attempting to participate in LEAPS

speculative call buying strategies, he should be more aggressive when rates and volatilities are low.

A few sample prices may help to demonstrate just how powerful the effects of rates and volatilities are, and how they can be a friend to the LEAPS call buyer. Suppose that one buys a 2-year LEAPS call at-the-money when the following situation exists:

XYZ: 100
January 2-year LEAPS call with strike of 100: 14
Short-term interest rates: 3%
Volatility: below average (historically)

For the purposes of demonstration, suppose that the current volatility is low for XYZ (historically) and that 3% is a low level for rates as well. If the stock moves up, there is no problem, because the LEAPS call will increase in price. But what if the stock drops or stays unchanged? Is all hope of a profit lost? Actually, no. If interest rates increase or the volatility that the calls trade at increases, we know the LEAPS call will increase in value as well. Thus, even though the direction in which the stock is moving may be unfavorable, it might still be possible to salvage one's investment. Table 25-3 shows where volatility would have to be or where short-term rates would have to go in order to keep the value of the LEAPS call at 14 even after the indicated amount of time had expired.

To demonstrate the use of this table, suppose the stock price were 100 (unchanged) after one month. If interest rates had risen to 3.4% from their original level of 3% during that time, the call would still be worth 14 even though one month had passed. Alternatively, if rates were the same, but volatility had increased by only 5% from its original level, then the call would also still be worth 14. Note that this means that volatility would have to increase only slightly (by $\frac{1}{20}$ th) from its original level, not by 5 percentage points.

TABLE 25-3.

Factors necessary for January 2-year LEAPS to be = 14.

Stock price	After 1 month	After 6 months
100 (unchanged)	$r = 3.4\%$ or $v + 5\%$	$r = 6\%$ or $v + 20\%$
95	$r = 6\%$ or $v + 20\%$	$r = 9.4\%$ or $v + 45\%$
90	$r = 8.5\%$ or $v + 45\%$	$r = 12.6\%$ or $v + 70\%$

Even if the stock were to drop to 90 and six months had passed, the LEAPS call holder would still be even if rates had risen to 12.6% (highly unlikely) or volatility had risen by 70%. It is often possible for volatilities to fluctuate to that extent in six months, but not likely for interest rates.

In fact, as interest rates go, only the top line of the table probably represents realistic interest rates; an increase of 0.4% in one month, or 3% in 6 months, is possible. The other lines, where the stock drops in price, probably require too large a jump in rates for rates alone to be able to salvage the call price. However, any increase in rates will be helpful. Volatility is another matter. It is often feasible for volatilities to change by as much as 50% from their previous level in a month, and certainly in six months. Hence, as has been stated before, the volatility factor is the more dominant one.

This table shows the effect of rising interest rates and volatilities on LEAPS calls. It would be beneficial to the LEAPS call owner and, of course, detrimental to the LEAPS call seller. This is clear evidence that one should be aware of the general level of rates and volatility before using LEAPS options in a strategy.

THE DELTA

The delta of an option is the amount by which the option price will change if the underlying stock changes in price by one point. In an earlier section of this chapter, comparing the differences between LEAPS and short-term calls, mention was made of delta. The subject is explored in more depth here because it is such an important concept, not only for option buyers, but for most strategic decisions as well.

Figure 25-5 depicts the deltas of two different options: 2-year LEAPS and 3-month equity options. Their terms are the same except for their expiration dates; striking price is 100, and volatility and interest rate assumptions are equal. The horizontal axis displays the stock price while the vertical axis shows the delta of the options.

Several relevant observations can be made. First, notice that the delta of the at-the-money LEAPS is very large, nearly 0.70. This means that the LEAPS call will move much more in line with the common stock than a comparable short-term equity option would. Very short-term at-the-money options have deltas of about $\frac{1}{2}$, while slightly longer-term ones have deltas ranging up to the 0.55 to 0.60 area. *What this implies is that the longer the life of an at-the-money option, the greater its delta.*

In addition, the figure shows that the deltas of the 3-month call and the 2-year LEAPS call are about equal when the options are approximately 5% in-the-money. If the options are more in-the-money than that, then the LEAPS call has a lower delta. This means that at- and out-of-the-money LEAPS will move more in line with the common stock than comparable short-term options will. Restated, the LEAPS calls will move

FIGURE 25-5.**Call delta comparison, 2-year LEAPS versus 3-month equity options.**

faster than the ordinary short-term equity calls unless both options are more than 5% in-the-money. Note that the movement referred to is in absolute terms in change of price, not in percentage terms.

The delta of the 2-year LEAPS does not change as dramatically when the stock moves as does the delta of the 3-month option (see Figure 25-5). Notice that the LEAPS curve is relatively flat on the chart, rising only slightly above horizontal. In contrast, the delta of the 3-month call is very low out-of-the-money and very large in-the-money. What this means to the call buyer is that the amount by which he can expect the LEAPS call to increase or decrease in price is somewhat stable. This can affect his choice of whether to buy the in-the-money call or the out-of-the-money call. With normal short-term options, he can expect the in-the-money call to much more closely mirror the movement in the stock, so he might be tempted to buy that call if he expects a small movement in the stock. With LEAPS, however, there is much less discrepancy in the amount of option price movement that will occur.

Example: XYZ is trading at 82. There are 3-month calls with strikes of 80 and 90, and there are 2-year LEAPS calls at those strikes as well. The following table summarizes the available information:

XYZ: 82		Date: January, 2002	
Option	Price	Delta	
April ('02) 80 call	4	$\frac{5}{8}$	
April ('02) 90 call	1	$\frac{1}{8}$	
January ('04) 80 LEAPS call	14	$\frac{3}{4}$	
January ('04) 90 LEAPS call	7	$\frac{1}{2}$	

Suppose the trader expects a 3-point move by the underlying common stock, from 82 to 85. If he were analyzing short-term calls, he would see his potential as a gain of $1\frac{5}{8}$ in the April 80 call versus a gain of $\frac{3}{8}$ in the April 90 call. Each of these gains is projected by multiplying the call's delta times 3 (the expected stock move, in points). Thus, there is a large difference between the expected gains from these two options, particularly after commissions are considered.

Now observe the LEAPS. The January 80 would increase by $2\frac{1}{4}$ while the January 90 would increase by $1\frac{1}{2}$ if XYZ moved higher by 3 points. This is not nearly as large a discrepancy as the short-term options had. Observe that the January 90 LEAPS sells for half the price of the January 80. These movements projected by the delta indicate that the January 90 LEAPS will move by a larger *percentage* than the January 80 and therefore would be the better buy.

PUT DELTAS

Many of the previous observations regarding deltas of LEAPS calls can be applied to LEAPS puts as well. However, Figure 25-5 changes a little when the following formula is applied. Recall that the relationship between put deltas and call deltas, except for deeply in-the-money puts, is:

$$\text{put delta} = \text{Call delta} - 1$$

This has the effect of *inverting* the relationships that have just been described. In other words, while the short-term calls didn't move as fast as the LEAPS, the *short-term puts move faster than the LEAPS puts in most cases*. Figure 25-6 shows the deltas of these options.

The vertical axis shows the puts' delta. Notice that out-of-the-money LEAPS puts and short-term equity puts don't behave very differently in terms of price change (bottom right-hand section of figure).

FIGURE 25-6.**Put delta comparison, 2-year LEAPS versus 3-month equity options.**

In-the-money puts (when the stock is below the striking price) move faster if they are shorter-term. This fact is accentuated even more when the puts are more deeply in-the-money. The left-hand side of the figure depicts this fact.

The LEAPS put delta curve is flat, just as the call delta curve was. Moreover, the delta is not very large anywhere across the figure. For example, at-the-money 2-year LEAPS puts move only about 30 cents for a one-point move in the underlying stock. *LEAPS put buyers who want to speculate on a stock's downward movement must realize that the leverage factor is not large; it takes approximately a 3-point move by the underlying common for an at-the-money LEAPS put to increase in value by one point.* Long-term puts don't mirror stock movement nearly as well as shorter-term puts do.

In summary, the option buyer who is considering buying LEAPS puts or calls as speculation should be aware of the different price action that LEAPS exhibit when compared to shorter-term options. Due to the large amount of time that LEAPS have remaining in their lives, the time decay of the LEAPS options is smaller. For this reason, the LEAPS option buyer doesn't need to be as precise in his timing. In general, LEAPS calls move faster when the underlying stock moves, and LEAPS puts move more slowly. Other than that, the general reasons for speculative option buying apply to LEAPS as well: leverage and limited risk.

SELLING LEAPS

Strategies involving selling LEAPS options do not differ substantially from those involving shorter-term options. The discussions in this section concentrate on the two major differences that sellers of LEAPS will notice. First, the slow rate of time decay of LEAPS options means that option writers who are used to sitting back and watching their written options waste away will not experience the same effect with LEAPS. Second, follow-up action for writing strategies usually depends on being able to buy back the written option when it has little or no time value premium remaining. Since LEAPS retain time value even when substantially in- or out-of-the-money, follow-up action involving LEAPS may involve the repurchase of substantial amounts of time value premium.

COVERED WRITING

LEAPS options can be sold against underlying stock just as short-term options can. No extra collateral or investment is required to do so. The resulting position is again one with limited profit potential, but enhanced profitability (as compared to stock ownership), if the underlying stock remains unchanged or falls. The maximum profit potential of the covered write is reached whenever the underlying stock is at or above the striking price of the written option at expiration.

The LEAPS covered writer takes in substantial premium, in terms of price, when he sells the long-term option. He should compare the return that he could make from the LEAPS write with returns that can be made from repeatedly writing shorter-term calls. Of course, there is no guarantee that he will actually be able to repeat the short-term writes during the longer life of the LEAPS.

As an aside, the strategist who is utilizing the *incremental return* concept of covered writing may find LEAPS call writing quite attractive. This is the strategy wherein he has a higher target price at which he would be willing to sell his common stock, and he writes calls along the way to earn an incremental return (see Chapter 2 for details). Since this type of writer is only concerned with absolute levels of premiums being brought into the account and not with things like return if exercised, he should utilize LEAPS calls if available, since the premiums are the largest available. Moreover, if the incremental return writer is currently in a short-term call and is going to be called away, he might roll into a LEAPS call in order to retain his stock and take in more premium.

The rest of this section discusses covered writing from the more normal viewpoint of the investor who buys stock and sells a call against it in order to attain a particular return.

Example: XYZ is selling at 50. The investor is considering a 500-share covered write and he is unsure whether to use the 6-month call or the 2-year LEAPS. The July 50 call sells

for 4 and has 6 months of life remaining; the January 50 LEAPS call sells for $8\frac{1}{2}$ and has 2 years of life. Further assume that XYZ pays a dividend of \$0.25 per quarter.

As was done in Chapter 2, the net required investment is calculated, then the return (if exercised) is computed, and finally the downside break-even point is determined.

Net Investment Required		
	July 50 call	January 50 LEAPS
Stock cost (500 shares @ 50)	\$25,000	\$25,000
Plus stock commission	+ 300	+ 300
Less option premiums received	- 2,000	- 4,250
Plus option commissions	+ 50	+ 100
Net cash investment	\$23,350	\$21,150

Obviously, the LEAPS covered writer has a smaller cash investment, since he is selling a more expensive call in his covered write. Note that the option premium is being applied against the net investment in either case, as is the normal custom when doing covered writing.

Now, using the net investment required, one can calculate the return (if exercised). That return assumes the stock is above the striking price of the written option at its expiration, and the stock is called away. The short-term writer would have collected two dividends of the common stock, while the LEAPS writer would have collected eight by expiration.

Return If Exercised		
	July 50 call	January 50 LEAPS
Stock sale (500 @ 50)	\$25,000	\$25,000
Less stock commission	- 300	- 300
Plus dividends earned until expiration	+ 250	+ 1,000
Less net investment	- 23,350	- 21,150
Net profit if exercised	\$ 1,600	\$ 4,550
Return if exercised (net profit/net investment)	6.9%	21.5%

The LEAPS writer has a much higher net return if exercised, again because he wrote a more expensive option to begin with. However, in order to fairly compare the two writes, one must annualize the returns. That is, the July 50 covered write made 6.9% in six months, so it could make twice that in one year, *if it can be duplicated six months*

from now. In a similar manner, the LEAPS covered writer can make 21.5% in two years if the stock is called away. However, on an annualized basis, he would make only half that amount.

Return If Exercised, Annualized	
July 50 call	January 50 LEAPS
13.8%	10.8%

Thus, on an annualized basis, the short-term write seems better. The shorter-term call will generally have a higher rate of return, annualized, than the LEAPS call. The problems with annualizing are discussed in the following text.

Finally, the downside break-even point can be computed for each write.

Downside Break-Even Calculation		
	July 50 call	January 50 LEAPS
Net investment	\$23,350	\$21,150
Less dividends received	— 250	— 1,000
Total stock cost to expiration	\$23,100	\$20,150
Divided by shares held (500), equals break-even price:	46.2	40.3

The larger premium of the LEAPS call that was written produces this dramatically lower break-even price for the LEAPS covered write.

Similar comparisons could be made for a covered write on margin if the investor is using a margin account. The steps above are the mechanical ones that a covered writer should go through in order to see how the short-term write compares to the longer-term LEAPS write. Analyzing them is often a less routine matter. It would seem that the short-term write is better if one uses the *annualized* rate of return. However, the annualized return is a somewhat subjective number that depends on several assumptions.

The first assumption is that one will be able to generate an equivalent return six months from now when the July 50 call expires worthless or the stock is called away. If the stock were relatively unchanged, the covered writer would have to sell a 6-month call for 4 points again six months from now. Or, if the stock were called away, he would have to invest in an equivalent situation elsewhere. Moreover, in order to reach the 2-year horizon offered by the LEAPS write, the 6-month return would have to be regenerated *three* more times (six months from now, one year from now, and a year and a half from

now). The covered writer cannot assume that such returns can be repeated with any certainty every six months.

The second assumption that was made when the annualized returns were calculated was that one-half the return if exercised on the LEAPS call would be made when one year had passed. But, as has been demonstrated repeatedly in this chapter, the time decay of an option is not linear. Therefore, one year from now, if XYZ were still at 50, the January 50 LEAPS call would not be selling for half its current price ($\frac{1}{2} \times 8\frac{1}{2} - 4\frac{1}{2}$). It would be selling for something more like 5.00, if all other factors remained unchanged. However, given the variability of LEAPS call premiums when interest rates, volatility, or dividend payouts change, it is extremely difficult to estimate the call price one year from now. Consequently, to say that the 21.5% 2-year return if exercised would be 10.8% after one year may well be a false statement.

Thus, the covered writer must make his decision based on what he knows. He knows that with the short-term July 50 write, if the stock is called away in six months, he will make 6.9%, period. If he opts for the longer term, he will make 21.5% if he is called away in two years. Which is better? The question can only be answered by each covered writer individually. One's attitude toward long-term investing will be a major factor in making the decision. If he thinks XYZ has good prospects for the long term, and he feels conservative returns will be below 10% for the next couple of years, then he would probably choose the LEAPS write. However, if he feels that there is a temporary expansion of option premium in the short-term XYZ calls that should be exploited, and he would not really want to be a long-term holder of the stock, then he would choose the short-term covered write.

Downside Protection. The actual downside break-even point might enter into one's thinking as well. A downside break-even point of 40.3 is available by using the LEAPS write, and that is a known quantity. No matter how far XYZ might fall, as long as it can recover to slightly over 40 by expiration two years from now, the investment will at least break even. A problem arises if XYZ falls to 40 quickly. If that happened, the LEAPS call would still have a significant amount of time value premium remaining on it. Thus, if the investor attempted to sell his stock at that time and buy back his call, he would have a loss, not a break-even situation.

The short-term write offers downside protection only to a stock price of 46.2. Of course, repeated writes of 6-month calls over the next 2 years would lower the break-even point below that level. The problem is that if XYZ declines and one is forced to keep selling 6-month calls every 6 months, he may be forced to use a lower striking price, thereby locking in a smaller profit (or possibly even a loss) if premium levels shrink. The concepts of rolling down are described in detail in Chapter 2.

A further word about rolling down may be in order here. Recall that rolling down means buying back the call that is currently written and selling another one with a lower

striking price. Such action *always* reduces the profitability of the overall position, although it may be necessary to prevent further downside losses if the common stock keeps declining. Now that LEAPS are available, the short-term writer faced with rolling down may look to the LEAPS as a means of bringing in a healthy premium even though he is rolling down. It is true that a large premium could be brought into the account. But remember that by doing so, one is committing himself to sell the stock at a lower price than he had originally intended. This is why the rolling down reduces the original profit potential. *If he rolls down into a LEAPS call, he is reducing his maximum profit potential for a longer period of time.* Consequently, one should not always roll down into an option with a longer maturity. Instead, he should carefully analyze whether he wants to be committed for an even longer time to a position in which the underlying common stock is declining.

To summarize, the large absolute premiums available in LEAPS calls may make a covered write of those calls seem unusually attractive. However, one should calculate the returns available and decide whether a short-term write might not serve his purpose as well. Even though the large LEAPS premium might reduce the initial investment to a mere pittance, be aware that this creates a great amount of leverage, and leverage can be a dangerous thing.

The large amount of downside protection offered by the LEAPS call is real, but if the stock falls quickly, there would definitely be a loss at the calculated downside break-even point. Finally, one cannot always roll down into a LEAPS call if trouble develops, because he will be committing himself for an even longer period of time to sell his stock at a lower price than he had originally intended.

“FREE” COVERED CALL WRITES

In Chapter 2, a strategy of writing expensive LEAPS options was briefly described. In this section, a more detailed analysis is offered. A certain type of covered call write, one in which the call is quite expensive, sometimes attracts traders looking for a “free ride.” To a certain extent, this strategy *is* something of a free ride. As you might imagine, though, there can be major problems.

The investment required for a covered call write on margin is 50% of the stock price, less the proceeds received from selling the call. In theory, it is possible for an option to sell for more than 50% of the stock cost. Thus in a margin account, a covered write could be established for “free.” Let’s discuss this in terms of two types of calls: the in-the-money call write and the out-of-the-money call write.

Out-of-the-Money Covered Call Write. This is the simplest way to approach the strategy. One may be able to find LEAPS options that are just slightly out-of-the-money, which sell for 50% of the stock price. Understandably, such a stock would be quite volatile.

Example: GOGO stock is selling for \$38 per share. GOGO has listed options, and a 2-year LEAPS call with a striking price of 40 is selling for \$19. The requirement for this covered write would be zero, although some commission costs would be involved. The debit balance would be 19 points per share, the amount the broker loans you on margin.

Certain brokerage firms might require some sort of minimum margin deposit, but technically there is no further requirement for this position. Of course, the leverage is infinite. Suppose one decided to buy 10,000 shares of GOGO and sell 100 calls, covered. *His risk is \$190,000 if the stock falls to zero!* That also happens to be the debit balance in his account. Thus, for a minimal investment, one could lose a fortune. In addition, if the stock begins to fall, one's broker is going to want maintenance margin. He probably wouldn't let the stock slip more than a couple of points before asking for margin. If one owns 10,000 shares and the broker wants two points maintenance margin, that means the margin call would be \$20,000.

The profits wouldn't be as big as they might at first seem. The maximum gross profit potential is \$210,000 if the 10,000 shares are called away at 40. The covered write makes 21 points on each share—the \$40 sale price less the original cost of \$19. However, one will have had to pay interest on the debit balance of \$190,000 for two years. At 10%, say, that's a total of \$38,000. There would also be commissions on the purchase and the sale.

In summary, this is a position with *tremendous*, even dangerous, leverage.

In-the-Money Covered Call Write. The situation is slightly different if the option is in-the-money to begin with. The above margin requirements actually don't quite accurately state the case for a margined covered call write. When a covered call is written against the stock, there is a catch: *Only 50% of the stock price or the strike price, whichever is less, is available for "release."* Thus, one will actually be required to put up more than 50% of the stock price to begin with.

Example: XYZ is trading at 50, and there is a 2-year LEAPS call with a strike price of 30, selling for 25 points. One might think that the requirement for a covered call write would be zero, since the call sells for 50% of the stock price. But that's *not* the case with in-the-money covered calls.

Margin requirement:

Buy stock: 50 points

Less option proceeds

–25

Less margin release*

–15*

Net requirement:

10 points

*50% of the strike price or 50% of stock price, whichever is less.

This position still has a lot of leverage: One invests 10 points in hopes of making 5, if the stock is called away at 30. One also would have to pay interest on the 15-point debit balance, of course, for the two-year duration of the position. Furthermore, should the stock fall below the strike price, the broker would begin to require maintenance margin.

Note that the above “formula” for the net requirement works equally well for the out-of-the-money covered call write, since 50% of the stock price is always less than 50% of the strike price in that case.

To summarize this “free ride” strategy: If one should decide to use this strategy, he must be extremely aware of the dangers of high leverage. One must not risk more money than he can afford to lose, regardless of how small the initial investment might be. Also, he must plan for some method of being able to make the margin payments along the way. Finally, the in-the-money alternative is probably better, because there is less probability that maintenance margin will be asked for.

SELLING UNCOVERED LEAPS

Uncovered option selling can be a viable strategy, especially if premiums are overpriced. LEAPS options may be sold uncovered with the same margin requirements as short-term options. Of course, the particular characteristics of the long-term option may either help or hinder the uncovered writer, depending on his objective.

Uncovered Put Selling. Naked put selling is addressed first because, as a strategy, it is equivalent to covered writing, and covered writing was just discussed. Two strategies are equivalent if they have the same profit picture at expiration. Naked put selling and covered call writing are equivalent because they have the profit picture depicted in Graph I, Appendix D. Both have limited upside profit potential and large loss exposure to the downside. In general, when two strategies are equivalent, one of the two has certain advantages over the other.

In this case, naked put selling is normally the more advantageous of the two because of the way margin requirements are set. One need not actually invest cash in the sale of a naked put; the margin requirement that is asked for may be satisfied with collateral. This means the naked put writer may use stocks, bonds, T-bills, or money market funds as collateral. Moreover, the actual amount of collateral that is required is less than the cash or margin investment required to buy stock and sell a call. This means that one could operate his portfolio normally—buying stock, then selling it and putting the proceeds in a Treasury bill or perhaps buying another stock—without disturbing his naked put position, as long as he maintained the collateral requirement.

Consequently, the *strategist* who is buying stock and selling calls should probably be selling naked puts instead. This does not apply to covered writers who are writing against existing stock or who are using the incremental return concept of covered writing, because stock ownership is part of their strategy. However, the strategist who is looking to take in premium to profit if the underlying stock remains relatively unchanged or rises, while having a modicum of downside protection (which is the definition of both naked put writing and covered writing), should be selling naked puts. As an example of this, consider the LEAPS covered write discussed previously.

Example: XYZ is selling at 50. The investor is debating between a 500-share covered write using 2-year LEAPS calls or selling five 2-year LEAPS puts. The January 50 LEAPS call sells for 8.50 and has two years of life, while the January 50 LEAPS put sells for 3.50. Further assume that XYZ pays a dividend of \$0.25 per quarter.

The net investment required for the covered write is calculated as it was before; assume that commissions are 3 cents per share for stock and \$5 for an option contract.

Net Investment Required—Covered Write	
Stock cost (500 shares @ 50)	\$25,000
Plus stock commission	+ 15
Less option premiums received	– 4,250
Plus option commissions	+ 25
Net cash investment	\$20,790

The collateral requirement for the naked put write is the same as that for any naked equity option: 20% of the stock price, plus the option price, less any out-of-the-money amount, with an absolute minimum requirement of 15% of the stock price.

Collateral Requirement—Naked Put	
20% of stock price ($.20 \times 500 \times \$50$)	\$5,000
Plus option premium	1,750
Less out-of-the-money amount	– 0
Total collateral requirement	\$6,750

Note that the actual premium received by the naked put seller is \$1,750 less commissions of \$25, or \$1,725. This net premium could be used to reduce the total collateral requirement.

Now one can compare the profitability of the two investments:

Return If Stock Over 50 at Expiration	
	Covered Write
Stock sale (500 @ 50)	\$25,000
Less stock commission	– 15
Plus dividends earned until expiration	+ 1,000
Less net investment	– 20,790
Net profit if exercised	\$ 5,195
	Naked Put Sale
Net put premium received	\$1,725
Dividends received	0
Net profit	\$1,725

Now the returns can be compared, if XYZ is over 50 at expiration of the LEAPS:

Return if XYZ over 50
(net profit/net investment)
Naked put sale: 25.6%
Covered write: 25.0%

The naked put write has a better rate of return, even before the following fact is considered. The strategist who is using the naked put write does not have to spend the \$6,750 collateral requirement in the form of cash. That money can be kept in a Treasury bill and earn interest over the two years that the put write is in place. Even if the T-bill were earning only 4% per year, that would increase the overall two-year return for the naked put sale by 8%, to 33.6%. This should make it obvious that *naked put selling is more strategically advantageous than covered call writing*.

Even so, one might rightfully wonder if LEAPS put selling is better than selling shorter-term equity puts. As was the case with covered call writing, the answer depends on what the investor is trying to accomplish. Short-term puts will not bring as much premium into the account, so when they expire, one will be forced to find another suitable put sale to replace it. On the other hand, the LEAPS put sale brings in a larger premium and one does not have to find a replacement until the longer-term LEAPS put expires. The negative aspect to selling the LEAPS puts is that time decay won't help much right away and, even if the stock moves higher (which is ostensibly good for the position), the put won't decline in price by a large amount, since the delta of the put is relatively small.

One other factor might enter in the decision regarding whether to use short-term puts or LEAPS puts. Some put writers are actually attempting to buy stock below the market price. That is, they would not mind being assigned on the put they sell, meaning that they would buy stock at a net cost of the striking price less the premium they received from the sale of the put. If they don't get assigned, they get to keep a profit equal to the premium they received when they first sold the put. Generally, a person would only sell puts in this manner on a stock that he had faith in, so that if he was assigned on the put, he would view that as a buying opportunity in the underlying stock. This strategy does not lend itself well to LEAPS. Since the LEAPS puts will carry a significant amount of time premium, there is little (if any) chance that the put writer will actually be assigned until the life of the put shortens substantially. This means that it is unlikely that the put writer will become a stock owner via assignment at any time in the near future. Consequently, if one is attempting to write puts in order to eventually buy the common stock when he is assigned, he would be better served to write shorter-term puts.

UNCOVERED CALL SELLING

There are very few differences between using LEAPS for naked call selling and using shorter-term calls, except for the ones that have been discussed already with regard to selling uncovered LEAPS: Time value decay occurs more slowly and, if the stock rallies and the naked calls have to be covered, the call writer will normally be paying more time premium than he is used to when he covers the call. Of course, the reason that one is engaged in naked call writing might shed some more light on the use of LEAPS for that purpose.

The overriding reason that most strategists sell naked calls is to collect the time premium before the stock can rise above the striking price. These strategists generally have an opinion about the stock's direction, believing that it is perhaps trapped in a trading range or even headed lower over the short term. This strategy does not lend itself well to using LEAPS, since it would be difficult to project that the stock would remain below the strike for so long a period of time.

Short LEAPS Instead of Short Stock. Another reason that naked calls are sold is as a strategy akin to shorting the common stock. In this case, in-the-money calls are sold. The advantages are threefold:

1. The amount of collateral required to sell the call is less than that required to sell stock short.
2. One does not have to borrow an option in order to sell it short, although one must borrow common stock in order to sell it short.
3. An uptick is not required to sell the option, but one is required in order to sell stock short.

For these reasons, one might opt to sell an in-the-money call instead of shorting stock.

The profit potentials of the two strategies are different. The short seller of stock has a very large profit potential if the stock declines substantially, while the seller of an in-the-money call can collect only the call premium no matter how far the stock drops. Moreover, the call's price decline will slow as the stock nears the strike. Another way to express this is to say that the delta of the call shrinks from a number close to 1 (which means the call mirrors stock movements closely) to something more like 0.50 at the strike (which means that the call is only declining half as quickly as the stock).

Another problem that may occur for the call seller is early assignment, a topic that is addressed shortly. *One should not attempt this strategy if the underlying stock is not borrowable for ordinary short sales.* If the underlying stock is not available for borrowing, it generally means that extraneous forces are at work; perhaps there is a tender offer or exchange offer going on, or some form of convertible arbitrage is taking place. In any case, if the underlying stock is not borrowable, one should not be deluded into thinking that he can sell an in-the-money call instead and have a worry-free position. In these cases, the call will normally have little or no time premium and may be subject to early assignment. If such assignment does occur, the strategist will become short the stock and, since it is not borrowable, will have to cover the stock. At the least, he will cost himself some commissions by this unprofitable strategy; and at worst, he will have to pay a higher price to buy back the stock as well.

LEAPS calls may help to alleviate this problem. Since they are such long-term calls, they are likely to have some time value premium in them. In-the-money calls that have time value premium are not normally assigned. As an alternative to shorting a stock that is not borrowable, one might try to sell an in-the-money LEAPS call, but *only if it has time value premium remaining*. Just because the call has a long time remaining until expiration does not mean that it must have time value premium, as will be seen in the following discussion. Finally, if one does sell the LEAPS call, he must realize that if the stock drops, the LEAPS call will not follow it completely. As the stock nears the strike, the amount of time value premium will build up to an even greater level in the LEAPS. Still, the naked call seller would make some profit in that case, and it presents a better alternative than not being able to sell the stock short at all.

Early Assignment. An American-style option is one that can be exercised at any time during its life. All listed equity options, LEAPS included, are of this variety. Thus, any in-the-money option that has been sold may become subject to early assignment. The clue to whether early assignment is imminent is whether there is time value premium in the option. If the option has no time value premium—in other words, it is trading at parity or at a discount—then assignment may be close at hand. The option writer who does not want to be assigned would want to cover the option when it no longer carries time premium.

LEAPS may be subject to early assignment as well. It is possible, albeit far less likely, that a long-term option would lose all of its time value premium and therefore be subject to early assignment. This would certainly happen if the underlying stock were being taken over and a tender offer were coming to fruition. However, it may also occur because of an impending dividend payment, or more specifically, because the stock is about to go ex-dividend. Recall that the call owner, LEAPS calls included, is not entitled to any dividends paid by the underlying stock. So if the call owner wants the dividend, he exercises his call on the day before the stock goes ex-dividend. This makes him an owner of the common stock just in the nick of time to get the dividend.

What economic factors motivate him to exercise the call? If there is any time value premium at all in the call, the call holder would be better off selling the call in the open market and then purchasing the stock in the open market as well. In this manner, he would still get the dividend, but he would get a better price for his call when he sold it. If, however, there is no time value premium in the call, he does not have to bother with two transactions in the open market; he merely exercises his call in order to buy stock.

All well and good, but what makes the call sell at parity before expiration? It has to do with the arbitrage that is available for any call option. In this case, the arbitrage is not the simple discount arbitrage that was discussed in Chapter 1 when this topic was covered. Rather, it is a more complicated form that is discussed in greater detail in Chapter 28. Suffice it to say that if the dividend is larger than the interest that can be earned from a credit balance equal to the striking price, then the time value premium will disappear from the call.

Example: XYZ is a \$30 stock and about to go ex-dividend 50 cents. The prevailing short-term interest rate is 5% and there are LEAPS with a striking price of 20.

A 50-cent quarterly dividend on a striking price of 20 is an annual dividend rate (on the strike) of 10%. Since short-term rates are much lower than that, arbitrageurs economically cannot pay out 10% for dividends and earn 5% for their credit balances.

In this situation, the LEAPS call would lose its time value premium and would be a candidate for early exercise when the stock goes ex-dividend.

In actual practice, the situation is more complicated than this, because the price of the puts comes into play; but this example shows the general reasoning that the arbitrageur must go through.

Certain arbitrageurs construct positions that allow them to earn interest on a credit balance equal to the striking price of the call. This position involves being short the underlying stock and being long the call. Thus, when the stock goes ex-dividend, the arbitrageur will owe the dividend. If, however, the amount of the dividend is more than he

will earn in interest from his credit balance, he will merely exercise his call to cover his short stock. This action will prevent him from having to pay out the dividend.

The arbitrageur's exercise of the call means that someone is going to be assigned. If you are a writer of the call, it could be you. It is not important to understand the arbitrage completely; its effect will be reflected in the marketplace in the form of a call trading at parity or a discount. *Thus, even a LEAPS call with a substantial amount of time remaining may be assigned if it is trading at parity.*

STRADDLE SELLING

Straddle selling is equivalent to ratio writing and is a strategy whereby one attempts to sell (overpriced) options in order to produce a range of stock prices within which the option seller can profit. The strategy often involves follow-up action as the stock moves around, and the strategist feels that he must adjust his position in order to prevent large losses. LEAPS puts and calls might be used for this strategy. However, their long-term nature is often not conducive to the aims of straddle selling.

First, consider the effect of time decay. One might normally sell a three-month straddle. If the stock "behaves" and is relatively unchanged after two months have passed, the straddle seller could reasonably expect to have a profit of about 40% of the original straddle price. However, if one had sold a 2-year LEAPS straddle, and the stock were relatively unchanged after two months, he would only have a profit of about 7% of the original sale price. This should not be surprising in light of what has been demonstrated about the decaying of long-term options. It should make the straddle seller somewhat leery of using LEAPS, however, unless he truly thinks the options are overpriced.

Second, consider follow-up action. Recall that in Chapter 20, it was shown that the bane of the straddle seller was the whipsaw. A whipsaw occurs when one makes a follow-up protective action on one side (for instance, he does something bullish because the underlying stock is rising and the short calls are losing money), only to have the stock reverse and come crashing back down. Obviously, the more time left until expiration, the more likely it is that a whipsaw will occur after any follow-up action, and the more expensive it will be, since there will be a lot of time value premium left in the options that are being repurchased. This makes LEAPS straddle selling less than attractive.

LEAPS straddles may look expensive because of their large absolute price, and therefore may appear to be attractive straddle sale candidates. However, the price is often justified, and the seller of LEAPS straddles will be fighting sudden stock movements without getting much benefit from the passage of time. The best time to sell LEAPS straddles is when short-term rates are high and volatilities are high as well (i.e., the options are overpriced). At least, in those cases, the seller will derive some real benefit if rates or volatilities should drop.

SPREADS USING LEAPS

Any of the spread strategies previously discussed can be implemented with LEAPS as well, if one desires. The margin requirements are the same for LEAPS spreads as they are for ordinary equity option spreads. One general category of spread lends itself well to using LEAPS: that of buying a longer-term option and selling a short-term one. Calendar spreads, as well as diagonal spreads, fall into that category.

The combinations are myriad, but the reasoning is the same. One wants to own the option that is not so subject to time decay, while simultaneously selling the option that is quite subject to time decay. Of course, since LEAPS are long-term and therefore expensive, one is generally taking on a large debit in such a spread and may have substantial risk if the stock performs adversely. Other risks may be present as well. As a means of demonstrating these facts, let us consider a simple bull spread using calls.

Example: The following prices exist in the month of January:

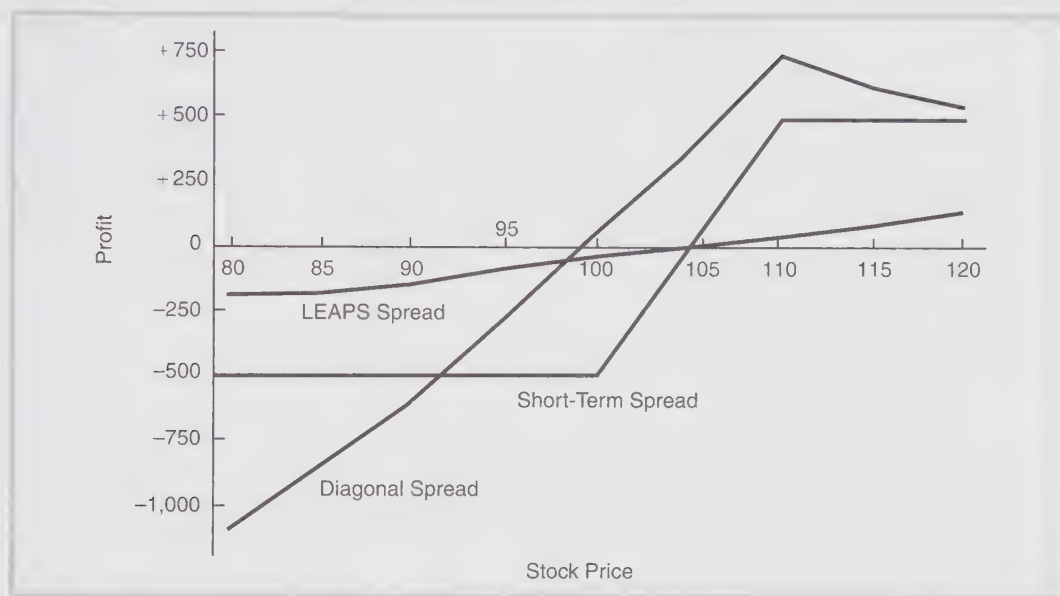
XYZ: 105
April 100 call: 10.50
April 110 call: 5.50
January (2-year) 100 call: 26
January (2-year) 110 call: 21.50

An investor is considering a bull spread in XYZ and is unsure about whether to use the short-term calls, the LEAPS calls, or a mixture. These are his choices:

Short-term bull spread:	Buy April 100 @ 10.50 Sell April 110 @ 5.50 Net Debit: \$500
Diagonal bull spread:	Buy January LEAPS 100 @ 26 Sell April 110 @ 5.50 Net Debit: \$2,050
LEAPS bull spread:	Buy January LEAPS 100 @ 26 Sell January LEAPS 110 @ 21.50 Net Debit: \$450

Notice that the debit paid for the LEAPS spread is slightly less than that of the short-term bull spread. This means that they have approximately the same profit potential at

FIGURE 25-7.
Bull spread comparison at April expiration.



their respective expiration dates. However, the strategist is more concerned with how these compare directly with each other. The obvious point in time to make this comparison is when the short-term options expire.

Figure 25-7 shows the profitability of these three positions at April expiration. It was assumed that all of the following were the same in April as they had been in January: volatility, short-term rates, and dividend payout.

Note that the short-term bull spread has the familiar profit graph from Chapter 7, making its maximum profit over 110 and taking its maximum loss below 100. (See Table 25-4.)

The LEAPS spread doesn't generate much of either a profit or a loss in only three months' time. Even if XYZ rises to 120, the LEAPS bull spread will have only a \$150 profit. Conversely, if XYZ falls to 80, the spread loses only about \$200. This price action is very typical for long-term bull spreads when both options have a significant amount of time premium remaining in them.

The diagonal spread is different, however. Typically, the maximum profit potential of a bull spread is the difference in the strikes less the initial debit paid. For this diagonal spread, that would be \$1,000 minus \$2,050, a loss! Obviously, this simple formula is not applicable to diagonal spreads, because the purchased option still has time value premium when the written option expires. The profit graph shows that indeed the diagonal spread is the most bullish of the three. It makes its best profit at the strike of the written option—a

TABLE 25-4.
Bull spread comparison at April expiration.

Stock Price	Short-Term	Diagonal	LEAPS
80	-500	-1,100	-200
90	-500	- 600	-150
100	-500	50	- 25
110	500	750	50
120	500	550	150
140	500	150	250
160	500	- 50	350
180	500	- 350	450

standard procedure for any spread—and that profit it is greater than either of the other two spreads *at April expiration* (under the significant assumption that volatility and interest rates are unchanged). If XYZ trades higher than 110, the diagonal spread will lose some of its profit; in fact, if XYZ were to trade at a very high price, the diagonal spread would actually have a loss (see Table 25-4). Whenever the purchased LEAPS call loses its time value premium, the diagonal spread will not perform as well.

If the common stock drops in price, the diagonal spread has the greatest risk in dollar terms but not in percentage terms, because it has the largest initial debit. If XYZ falls to 80 in three months, the spread will lose about \$1,100, just over half the initial \$2,050 debit. Obviously, the short-term spread would have lost 100% of its initial debit, which is only \$500, at that same point in time.

The diagonal spread presents an opportunity to earn more money if the underlying common is near the strike of the written option when the written option expires. However, if the common moves a great deal in either direction, the diagonal spread is the worst of the three. This means that the diagonal spread strategy is a neutral strategy: One wants the underlying common to remain near the written strike until the near-term option expires. This is a true statement even if the diagonal spread is under the guise of a bullish spread, as in the previous example.

Many traders are fond of buying LEAPS and selling an out-of-the-money near-term call as a hedge. Be careful about doing this. If the underlying common rises too fast and/or interest rates fall and/or volatility decreases, this could be a poor strategy. There is really nothing quite as psychologically damaging as being right about the stock, but being in the wrong option strategy and therefore losing money. Consider the above examples. Ostensibly, the spreader was bullish on XYZ; that's why he chose bull spreads. If XYZ

became a wildly bullish stock and rose from 100 to 180 in three months, the diagonal spreader would have lost money. He couldn't have been happy—no one would be. This is something to keep in mind when diagonalizing a LEAPS spread.

The deltas of the options involved in the spread will give one a good clue as to how it going to perform. Recall that a short-term, in-the-money option acquires a rather high delta, especially as expiration draws nigh. However, an in-the-money LEAPS call will *not* have an extremely high delta, because of the vast amount of time remaining. Thus, one is short an option with a high delta and long an option with a smaller delta. These deltas indicate that one is going to lose money if the underlying stock rises in price. Consider the following situation:

XYZ Stock, 120:	
Call	Position Delta
Long 1 January LEAPS 100 call:	0.70
Short 1 April 110 call:	-0.90

At this point, if XYZ rises in price by 1 point, the spread can be expected to lose 20 cents, since the delta of the short option is 0.20 greater than the delta of the long option.

This phenomenon has ramifications for the diagonal spreader of LEAPS. If the two strike prices of the spread are too close together, it may actually be possible to construct a bull spread that *loses* money on the upside. That would be very difficult for most traders to accept. In the above example, as depicted in Table 25-4, that's what happens. One way around this is to widen the strike prices out so that there is at least *some* profit potential, even if the stock rises dramatically. That may be difficult to do and still be able to sell the short-term option for any meaningful amount of premium.

Note that a diagonal spread could even be considered as a substitute for a covered write in some special cases. It was shown that a LEAPS call can sometimes be used as a substitute for the common stock, with the investor placing the difference between the cost of the LEAPS call and the cost of the stock in the bank (or in T-bills). Suppose that an investor is a covered writer, buying stock and selling relatively short-term calls against it. If that investor were to make a LEAPS call substitution for his stock, he would then have a diagonal bull spread. Such a diagonal spread would probably have less risk than the one described above, since the investor presumably chose the LEAPS substitution because it was "cheap." Still, the potential pitfalls of the diagonal bull spread would apply to this situation as well. Thus, if one is a covered writer, this does not necessarily mean that he can substitute LEAPS calls for the long stock without taking care. The resulting position may not resemble a covered write as much as he thought it would.

The “bottom line” is that if one pays a debit greater than the difference in the strike prices, he may eventually lose money if the stock rises far enough to virtually eliminate the time value premium of both options. This comes into play also if one rolls his options *down* if the underlying stock declines. Eventually, by doing so, he may *invert* the strikes—i.e., the striking price of the written option is lower than the striking price of the option that is owned. In *that* case, he will have locked in a loss if the overall *credit* he has received is less than the difference in the strikes—a quite likely event. So, for those who think this strategy is akin to a guaranteed profit, think again. It most certainly is not.

Backspreads. LEAPS may be applied to other popular forms of diagonal spreads, such as the one in which in-the-money, near-term options are sold, and a greater quantity of longer-term (LEAPS) at- or out-of-the money calls are bought. (This was referred to as a diagonal backspread in Chapter 14.) This is an excellent strategy, and a LEAPS may be used as the long option in the spread. Recall that the object of the spread is for the stock to be volatile, particularly to the upside if calls are used. If that doesn’t happen, and the stock declines instead, at least the premium captured from the in-the-money sale will be a gain to offset against the loss suffered on the longer-term calls that were purchased. The strategy can be established with puts as well, in which case the spreader would want the underlying stock to fall dramatically while the spread was in place.

Without going into as much detail as in the examples above, the diagonal backspreader should realize that he is going to have a significant debit in the spread and could lose a significant portion of it should the underlying stock fall a great deal in price. To the upside, his LEAPS calls will retain some time value premium and will move quite closely with the underlying common stock. Thus, he does not have to buy as many LEAPS as he might think in order to have a neutral spread.

Example: XYZ is at 105 and a spreader wants to establish a backspread. Recall that the quantity of options to use in a neutral strategy is determined by dividing the deltas of the two options. Assume the following prices and deltas exist:

XYZ: 105 in January		
Option	Price	Delta
April 100 call	8	0.75
July 110 call	5	0.50
January (2-year) LEAPS 100 call	15	0.60

Two backspreads are available with these options. In the first, one would sell the April 100 calls and buy the July 110 calls. He would be selling 3-month calls and buying

6-month calls. The neutral ratio is 0.75/0.50 or 3 to 2; that is, 3 calls are to be bought for every 2 sold. Thus, a neutral spread would be:

Buy 6 July 110 calls
Sell 4 April 100 calls

As a second alternative, he might use the LEAPS as the long side of the spread; he would still sell the April 100 calls as the short side of the spread. In this case, his neutral ratio would be 0.75/0.60, or 5 to 4. The resulting neutral spread would be:

Buy 5 January LEAPS 110 calls
Sell 4 April 100 calls

Thus, a neutral backspread involving LEAPS requires buying fewer calls than a neutral backspread involving a 6-month option on the long side. This is because the delta of the LEAPS call is larger. The significant point here is that, because of the time value retention of the LEAPS call, even when the stock moves higher, it is not necessary to buy as many. If one does not use the deltas, but merely figures that 3 to 2 is a good ratio for any diagonal backspread, then he will be overly bullish if he uses LEAPS. That could cost him if the underlying stock declines.

Calendar Spreads. LEAPS may also be used in calendar spreads—spreads in which the striking price of the longer-term option purchased and the shorter-term option sold are the same. The calendar spread is a neutral strategy, wherein the spreader wants the underlying stock to be as close as possible to the striking price when the near-term option expires. A calendar spread has risk if the stock moves too far away from the striking price (see Chapter 9 and 22). Purchasing a LEAPS call increases that risk in terms of dollars, not percentage, because of the larger debit that one must spend for the spread.

Simplistically, calendar spreads are established with equal quantities of options bought and sold. This is often not a neutral strategy in the true sense. As was shown in Chapter 9 on call calendar spreads, one may want to use the deltas of the two options to establish a truly neutral calendar spread, particularly if the stock is not initially right at the striking price. Out-of-the-money, one would sell more calls than he is buying. Conversely, in-the-money, one would buy more calls than he is selling. Both strategies statistically have merit and are attractive. When using LEAPS deltas to construct the neutral spread, one need generally buy fewer calls than he might think, because of the higher delta of a LEAPS call. This is the same phenomenon described in the previous example of a diagonal backspread.

SUMMARY

LEAPS are nothing more than long-term options. They are usable in a wide variety of strategies in the same way that any option would be. Their margin and investment requirements are similar to those of the more familiar equity options. Both LEAPS puts and calls are traded, and there is a secondary market for them as well.

There are certain differences between the prices of long-term options and those of shorter-term options, but the greatest is the relatively large effect that interest rates and dividends have on the price of long-term options. Increases in interest rates will cause LEAPS to increase in price, while increases in dividend payout will cause LEAPS calls to decrease in price and LEAPS puts to increase in price. As usual, volatility has a major effect on the price of an option, and long-term options are no exception. Even small changes in the volatility of the underlying common stock can cause large price differences in a two-year option. The rate of decay due to time is much smaller for long-term options. Finally, the deltas of LEAPS calls are larger than those of short-term calls; conversely, the deltas of LEAPS puts are smaller.

Several common strategies lend themselves well to the usage of LEAPS. A LEAPS may be used as a stock substitute if the cash not invested in the stock is instead deposited in a CD or T-bill. LEAPS puts can be bought as protection for common stock. Speculative option buyers will appreciate the low rate of time decay of LEAPS. LEAPS calls can be written against common stock, thereby creating a covered write, although the sale of naked LEAPS puts is probably a better strategy in most cases. Spread strategies with LEAPS may be viable as well, but the spreader should carefully consider the ramifications of buying a long-term option and selling a shorter-term one against it. If the underlying stock moves a great distance quickly, the spread strategy may not perform as expected.

Overall, long-term options are not very different from the shorter-term options to which traders and investors have become accustomed. Once these investors become familiar with the way these long-term options are affected by the various factors that determine the price of an option, they will consider the use of long-term options as an integral part of a strategic arsenal.